Sarbartha Sarkar

Email: sarbarthasarkar2611@gmail.com



Education

- **Barasat Indira Gandhi Memorial High School** — Kolkata, India
  Class 10 — Grade: 93% — *2021*
- **Barasat Indira Gandhi Memorial High School** — Kolkata, India
  Class 12 — Grade: 92.8% — *2023*
- **Indian Statistical Institute** — Bengaluru, India
  Bachelor of Mathematics — Grade: 1st Division — *July 2023 – May 2026*
- **Indian Statistical Institute** — Kolkata, India
  Master of Mathematics — *July 2026 – May 2028*

Experience

- **Instructor @ UDGRP 2024-25**: I took a Winter Reading Project in Measure Theory in Winter 2024, which was conducted by the Math Club. We discussed sequences and series of functions, general measure theory, Lebesgue measure and its properties. We covered Vitali's construction of non-measurable sets, measurable sets and measurable functions, Egorov's theorem, and ended with Lebesgue integration.(December '24)
- **Talk @ Math Club**: I gave a talk at the Math Club of ISI Bengaluru. In the talk I talked about Hausdorff outer measure and its various properties. Then we discussed Hausdorff dimension and its correlation with the usual notions of topological and Euclidean dimension, and finally ended by showing that the Hausdorff dimension of the middle-third Cantor set is $\frac{\log 2}{\log 3}$. (September '25)
- **Talk @ Math Club**: I gave a talk at the math club of ISI Kolkata. In the talk I tried to motivate about various fractal sets and how they fail under the usual geometric intuition. Then using a scaling argument, the definition of the s-dimensional Hausdorff outer measure was motivated. Then we discussed the definition of Hausdorff dimension. Then we finally concluded by briefly discussing about its connection with Geometric Measure Theory and how its tools are used to study Fractal Geometry.(October '26)

Projects

- **UDGRP 2023-24**: I completed a Winter Reading Project in Measure Theory conducted by the Math Club. Here, I read about different types of measures such as the Dirac measure, the Lebesgue measure and its properties, Vitali's construction of non-measurable sets, measurable sets and measurable functions, and ended with the Lebesgue integral.(December '23)
- **Research Intern @ IISER Berhampur**: Under Professor Dr. Senthil Raani K. S., I did a reading project on Geometric Measure Theory, where I read about Hausdorff measure and its properties. Then I read about some important covering lemmas. After that, I studied Hausdorff dimension and some of its properties. We then calculated the Hausdorff dimension of different constructions of the Cantor set, including showing that the Hausdorff dimension of the middle-third Cantor set is $\frac{\log 2}{\log 3}$. I also read that, for the general construction of the Cantor set, the Hausdorff measure is 1. Finally, the project ended with studying the density of a set at a point in $\mathbb{R}^n$.(June '25)
- **Workshop @ IISC**: Attended a workshop on *Graphs, Spectra and Convex Geometry* at IISc. Read about the applications of Graph Theory in various fields, including Convex Geometry, under the guidance of Professor Dr. Rekha R. Thomas and Professor Dr. Arvind Ayyer.(July '25)
- **Reading project @ ISI Bengaluru**: Under Professor Dr. Haripada Sau of IISER Pune, I did a reading project on Geometric Measure Theory, particularly continuing my previous studies in the subject by exploring its connections with Functional Analysis. Here, we discussed the definitions of Hausdorff measure and Hausdorff dimension, some important covering arguments, the Vitali Covering Theorem, and the relation between the n-dimensional Lebesgue outer measure and the Hausdorff outer measure.(May '26-July '26)
- **Reading project in Geometric Measure Theory**: Reading project under Prof. Malabika Pramanik (University of British Columbia, Canada), continuing my earlier study of Geometric Measure Theory. Studied techniques for computing the Hausdorff dimension of fractals, namely the mass distribution principle, potential-theoretic methods and Fourier-analytic methods. Analysed how the Hausdorff dimension of various classes of Cantor sets depends on the scaling of their spacing conditions. (July '26–Present)

Achievements

- Rank 13 at Presidency University Bachelor Degree Entrance Test 2023.

Extracurricular Activities

- **Math club, ISI Kolkata**: Core committee member, organizing student talks and lecture series. Managed the drafting and sending of outreach emails inviting external speakers and mathematicians. (August '26–Present)

Independent Study Road Map

The following road map summarizes my independent study, in measure theory and fractal geometry, indicating the progression of topics and principal references that have shaped my current research interests.

